

Joint Matrix Completion and Randomized SVD Channel Estimation of the mmWave Massive MIMO

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Abstract—Beamforming technology is usually applied to millimeter wave(mmWave) massive MIMO systems to improve transmission gain, spectrum efficiency, and anti-interference ability. The effective acquisition of channel state information(CSI) is necessary for beamforming technology. However, the low signal-to-noise(SNR) environment of beamforming and the high variability of the mmWave channel make it challenging to acquire CSI. Recently, some researches on the sparsity of the mmWave channel in the angular domain have been carried out. And it's proved that the mmWave channel exhibits a joint sparse and low-rank structure where the rank is far smaller than the sparsity level of the channel. But numbers of training symbols are still needed in these works to achieve satisfactory performance. In this work, we propose a novel channel estimation algorithm based on matrix completion which jointly utilizes the sparsity and low-rank characteristics of the mmWave channel, and the randomized singular value decomposition(SVD) is introduced to accelerate the process of the algorithm. Simulation results demonstrate that the proposed channel estimation algorithm can show more accurate performance under the condition of shorter training sequences and a low SNR environment in the mmWave massive MIMO systems.

Index Terms—massive MIMO, millimeter wave(mmWave), channel estimation, matrix completion, alternating direction multiplier method(ADMM), randomized SVD.

I. INTRODUCTION

Supporting 1000 fold greater capacity is one of the major needs of the fifth-generation(5G) compared to the long-term evolution(LTE). Massive multiple-input-multiple-output(MIMO) technology is considered as one of the promising ways to improve spectrum efficiency by introducing a large number of antennas at transmitter (TX) and receiver (RX) [1]. Hence, a higher signal frequency than the microwave frequency band to support the integration of more antennas is needed. The mmWave distributed in 30-300GHz with a shorter wavelength can meet people's needs. However, mmWave is easily absorbed by the atmosphere during transmission, and the rainfall attenuation is serious, which results in large path loss. Thus the beamforming technique is introduced to elevate the path gain in the mmWave MIMO systems [2]. The effective acquisition of CSI is essential for beamforming technology, but reliable CSI is difficult to obtain because of the large number of antennas and the high mmWave channel variability.

However, previous studies have shown that the wireless channel between the base station and the user presents a small angle expansion. The angle spread is small compared with the channel size, which makes the channel sparse in the virtual angle domain [3]. Based on the sparsity of the mmWave channel, the problem of the mmWave channel estimation can be regarded as the problem of sparse signal recovery, which can be solved by compressed sensing technologies, such as the orthogonal matching pursuit (OMP) algorithm [4]. These can initially reduce the use of training sequences. However, a large number of training symbols are still needed to achieve satisfactory estimating performance. In addition, the mmWave channel exhibits angular spread in the domains of the angle of arrival(AOA), angle of departure(AOD), and elevation, which can produce a useful low-rank structure [5]. Paper [6] proposed a two-stage method for mmWave time-varying channel estimation. But the two stages are independent.

The low-rank characteristic of the mmWave channel can be solved by matrix completion. The problem of matrix completion or missing value estimation is widely used in computer vision, pattern recognition, and machine learning [7]. Generally, the matrix we need to recover is of low rank, and a widely used method is to use singular value thresholding(SVT) algorithm to solve it based on the convex optimization characteristic of matrix kernel norm. However, the SVT algorithm will produce high computational complexity in the mmWave massive MIMO systems. Because it needs to calculate the singular values and corresponding singular vectors that exceed the threshold in each iteration step. This is also a deficiency of the ADMM algorithm [8].

In this work, in view of the above problems, we propose a novel channel estimation algorithm based on matrix completion and randomized SVD that jointly utilizes the low-rank and sparse characteristics of the mmWave channel. The simulation results show that, compared with the other techniques [4], [5], [8], [9], [10], the proposed algorithm shows more accurate performance under the condition of shorter training sequences and a low SNR environment in the mmWave massive MIMO systems.

II. PRELIMINARIES

A. Channel Model

We consider that a point-to-point uplink scenario and an $N_R \times N_T$ massive MIMO system which adopts a relatively simple analog beamforming with switches for channel estimation [11]. This structure can be easily extended to hybrid beamforming(HBF) architectures with low complexity.

During the training period($\forall t = 1, \dots, T$), the TX and RX activate one antenna randomly, which means that the precoding vector at the transmitter and the combining vector at the receiver have only one non-zero value, respectively. Suppose the channel matrix $\mathbf{H} \in C^{N_R \times N_T}$ remains static during the transmission of T unit power training symbols $s[t] \in C, \forall t = 1, \dots, T$. The combined signal at the receiver is expressed as $r[t] = \mathbf{z}^H(t)\mathbf{H}\mathbf{f}(t)s[t] + w[t], \forall t = 1, \dots, T$, where $\mathbf{f}(t) \in \{0, 1\}^{N_T \times 1}$ and $\mathbf{z}(t) \in \{0, 1\}^{N_R \times 1}$ denote the TX precoding and RX combining vectors, respectively, and $w[t]$ represents the zero-mean complex additive white Gaussian noise with variance.

We adopt the extended Saleh-Valenzuela model to describe the feature of the mmWave channel. Suppose that the antenna array is a uniform linear array(ULA), then the channel matrix $\mathbf{H}$ can be expressed as

$$\mathbf{H} = \sum_{l=1}^L \alpha_l \mathbf{a}_R(\theta_l) \mathbf{a}_T^H(\phi_l) \in C^{N_R \times N_T} \quad (1)$$

where L is the number of paths, α_l is the complex gain of the l th path. $\theta_l \in [0, 2\pi]$ and $\phi_l \in [0, 2\pi]$ are the associated azimuth AoA and azimuth AoD respectively. $\mathbf{a}_R \in C^{N_R \times 1}$ and $\mathbf{a}_T \in C^{N_T \times 1}$ are the array response vector associated with the BS and MS respectively, which can be written as

$$\mathbf{a}_T(\phi_l) = \left[1, e^{j\frac{d}{\lambda}2\pi \sin \phi_l}, \dots, e^{j(N_T-1)\frac{d}{\lambda}2\pi \sin \phi_l} \right] / \sqrt{N_T}$$

$$\mathbf{a}_R(\theta_l) = \left[1, e^{j\frac{d}{\lambda}2\pi \sin \theta_l}, \dots, e^{j(N_R-1)\frac{d}{\lambda}2\pi \sin \theta_l} \right] / \sqrt{N_R}$$

where d denotes the distance between neighboring antenna elements, λ is the signal wavelength.

In addition, when the base station is equipped with ULAs, the massive MIMO channel has an almost sparse expression under the discrete Fourier transform(DFT). Then $\mathbf{H}$ can be represented based on the beamspace model [12], which is defined as

$$\mathbf{H} = \mathbf{D}_R \mathbf{S} \mathbf{D}_T^H \quad (2)$$

where $\mathbf{D}_R \in C^{N_R \times N_R}$ and $\mathbf{D}_T \in C^{N_T \times N_T}$ are the normalized DFT matrices for ULA. It means that $\mathbf{D}_R^H \mathbf{D}_R = \mathbf{I}_{N_R}$ and $\mathbf{D}_T^H \mathbf{D}_T = \mathbf{I}_{N_T}$. $\mathbf{S} \in C^{N_R \times N_T}$ is a sparse matrix which contains a few virtual channel gains with high amplitude. Therefore, we can represent the sparse characteristics of the mmWave channel by means of matrix $\mathbf{S}$, which has great application in the next phase of matrix completion. The sparsity level of $\mathbf{S}$ depends on the angular discretization in the beamspace representation given by (2).

B. Randomized SVD Algorithms

The research on the operation of a random matrix is getting more popular in recent years. Randomized algorithms involve the same or less floating-point operations and have great advantages in solving linear least squares problems and low-rank matrix approximation. Since the largest singular values contain most of the information of the original matrix, we can use this method to avoid calculating all the eigenvalues of a matrix and reduce the amount of calculation.

We combine a basic randomized SVD (rSVD) algorithm [13] with the block krylov subspace iteration (BKI) scheme [14]. The main idea of this algorithm is to stack the subspaces generated by each iteration to generate a larger subspace. The algorithm rSVD-BKI is described in Alg.1.

Algorithm 1 rSVD-BKI

Input: $\mathbf{A} \in R^{m \times n}$, rank parameter k , power parameter p
Output: $\mathbf{U} \in R^{m \times k}$, $\mathbf{S} \in R^{k \times k}$, $\mathbf{V} \in R^{n \times k}$

- 1: $\Omega_0 = \text{randn}(n, k + s)$
- 2: $[\mathbf{G}_0, \sim] = \text{lu}(\mathbf{A}\Omega)$
- 3: **for** $i = 1, 2, \dots, p$ **do**
- 4: **if** $i < p$ **then**
- 5: $[\mathbf{G}_i, \sim] = \text{lu}(\mathbf{A}(\mathbf{A}^T \mathbf{G}_{i-1}))$
- 6: **end if**
- 7: **end for**
- 8: $\mathbf{G} = [\mathbf{G}_0, \mathbf{G}_1, \dots, \mathbf{G}_p]$
- 9: $\mathbf{Q} = \text{orth}(\mathbf{G})$
- 10: $\mathbf{B} = \mathbf{Q}^T \mathbf{A}$
- 11: $[\mathbf{V}, \mathbf{S}, \mathbf{U}] = \text{eigSVD}(\mathbf{B}^T)$
- 12: $\text{ind} = (k + s)(p + 1) - k + 1 : (k + s)(p + 1)$
- 13: $\mathbf{U} = \mathbf{Q}\mathbf{U}(:, \text{ind}), \mathbf{S} = \mathbf{S}(\text{ind}, \text{ind}), \mathbf{V} = \mathbf{V}(:, \text{ind})$

In Alg.1, Ω_0 is a Gaussian sampling matrix, $\mathbf{Q}$ is the orthogonal basis of subspace. By introducing matrix $\mathbf{B} = \mathbf{Q}^T \mathbf{A}$, we have the approximation $\mathbf{A} \approx \mathbf{Q}\mathbf{B} = \mathbf{Q}\mathbf{Q}^T \mathbf{A}$ [15]. The parameter s in Alg.1 is an oversampling factor, which can make the number of columns of the random matrix Ω_0 exceed k , so as to obtain higher accuracy. It usually takes a small integer, for example, 5. The orthonormalization operation "orth()" can be achieved by a call to a packaged QR factorization.

And the SVD algorithm based on eigendecomposition is described in Alg.2. Suppose $\mathbf{A}$ is an $m \times n$ matrix, it can be decomposed as $\mathbf{A} = \mathbf{U}(:, 1 : n) \tilde{\mathbf{S}} \mathbf{V}^T$ when $m \geq n$, where $\tilde{\mathbf{S}}$ is a diagonal matrix with the first n singular values of $\mathbf{A}$. Then according the Alg.2, $\mathbf{B} = \mathbf{A}^T \mathbf{A} = \mathbf{V} \tilde{\mathbf{S}}^2 \mathbf{V}^T$. After the Step 2 of Alg.2, we get $\mathbf{D} = \tilde{\mathbf{S}}^2$, where $\mathbf{V}$ is the right singular vector matrix of $\mathbf{A}$. Hence, $\mathbf{S} = \text{sqrt}(\mathbf{D}) = \tilde{\mathbf{S}}$, and $\mathbf{U} = \mathbf{A}\mathbf{V}\mathbf{S}^{-1} = \mathbf{A}\mathbf{V}\tilde{\mathbf{S}}^{-1} = \mathbf{U}(:, 1 : n)$. Since $m \geq n$, dimension of $\mathbf{B}$ is $n \times n$, the eigenvalue decomposition of $\mathbf{B}$ is more efficient.

Algorithm 2 eigSVD**Input:** $\mathbf{A} \in \mathbf{R}^{m \times n}$ ($m \geq n$)**Output:** $\mathbf{U} \in \mathbf{R}^{m \times n}, \mathbf{S} \in \mathbf{R}^{n \times n}, \mathbf{V} \in \mathbf{R}^{n \times n}$

- 1: $\mathbf{B} = \mathbf{A}^T \mathbf{A}$
- 2: $[\mathbf{V}, \mathbf{D}] = \text{eig}(\mathbf{B})$
- 3: $\mathbf{S} = \text{sqrt}(\mathbf{D})$
- 4: $\mathbf{U} = \mathbf{A} \mathbf{V} \mathbf{S}^{-1}$

III. PROPOSED mmWAVE MIMO CHANNEL ESTIMATION

A. Problem Formulation

Matrix completion theory takes the known low-rank matrix elements as constraints and restores the original matrix elements by solving the optimization problem of rank minimization [9]. According to the mathematical problem that the norm of matrix kernel is the optimal convex substitution of rank function, we can establish the recovery problem of rank minimization as

$$\begin{aligned} \min \quad & \|\mathbf{H}\|_* \\ \text{s.t.} \quad & \rho_{\Omega_1}(\mathbf{H}) = \rho_{\Omega_1}(\mathbf{D}) \end{aligned} \quad (3)$$

where $\mathbf{H} \in \mathbf{C}^{M \times N}$ is a low-rank matrix, which satisfies $\text{rank}(\mathbf{H}) \ll \min\{M, N\}$. Ω_1 is the set of observation samples. $\rho_{\Omega_1}(\cdot) : \mathbf{C}^{M \times N} \rightarrow \mathbf{C}^{M \times N}$ denotes the projection matrix [16].

We take the beamspace representation of $\mathbf{H}$ given by (2) as (3)'s side information. Then we formulate the following constrained optimization problem (OP) as

$$\begin{aligned} \min_{\mathbf{H}, \mathbf{S}} \quad & \tau_H \|\mathbf{H}\|_* + \tau_S \|\mathbf{S}\|_1 \\ \text{s.t.} \quad & \Omega^\circ \mathbf{H} = \mathbf{H}_\Omega \text{ and } \mathbf{H} = \mathbf{D}_R \mathbf{S} \mathbf{D}_T^H \end{aligned} \quad (4)$$

where $\Omega \in \{0, 1\}^{N_R \times N_T}$, it denotes the sampling matrix in the training period. Because there is only one transceiver antenna working at each time, the matrix has only T non-zero values. $\mathbf{H}_\Omega$ is the observation matrix of $\mathbf{H}$. It's obvious that the nonzero values in $\mathbf{H}_\Omega$ are arranged in the same way as matrix Ω . $\|\mathbf{S}\|_1 \triangleq \max_{1 \leq j \leq N_T} \sum_{i=1}^{N_R} |\mathbf{S}_{i,j}|$. The weight factors τ_H and τ_S are greater than 0, and their values are related to the number of mmWave channel propagation paths.

B. Proposed MIMO Channel Estimation Algorithm

The optimization problem(OP) of (4) is a two-objective convex problem, thus it has a global optimal solution. We introduce two auxiliary matrix variables $\mathbf{Y} \in \mathbf{C}^{N_R \times N_T}$ and $\mathbf{C} \triangleq \mathbf{Y} - \mathbf{D}_R \mathbf{S} \mathbf{D}_T^H$ to reformulate (4) to

$$\begin{aligned} \min_{\mathbf{H}, \mathbf{S}} \quad & \tau_H \|\mathbf{H}\|_* + \tau_S \|\mathbf{S}\|_1 + \frac{1}{2} \|\mathbf{C}\|_F^2 + \frac{1}{2} \|\Omega^\circ \mathbf{Y} - \mathbf{H}_\Omega\|_F^2 \\ \text{s.t.} \quad & \mathbf{H} = \mathbf{Y} \text{ and } \mathbf{C} \triangleq \mathbf{Y} - \mathbf{D}_R \mathbf{S} \mathbf{D}_T^H \end{aligned} \quad (5)$$

Now the OP in (5) has separate blocks of variables, which enables ADMM utilization, and the augmented Lagrangian function of (5) is given by

$$\begin{aligned} L_1(\mathbf{H}, \mathbf{Y}, \mathbf{S}, \mathbf{C}, \mathbf{Z}_1, \mathbf{Z}_2) \triangleq & \tau_H \|\mathbf{H}\|_* + \tau_S \|\mathbf{S}\|_1 + \frac{1}{2} \|\mathbf{C}\|_F^2 \\ & + \frac{1}{2} \|\Omega^\circ \mathbf{Y} - \mathbf{H}_\Omega\|_F^2 + \text{tr}(\mathbf{Z}_1^H (\mathbf{H} - \mathbf{Y})) + \frac{\rho}{2} \|\mathbf{H} - \mathbf{Y}\|_F^2 \\ & + \text{tr}(\mathbf{Z}_2^H (\mathbf{Y} - \mathbf{D}_R \mathbf{S} \mathbf{D}_T^H - \mathbf{C})) + \frac{\rho}{2} \|\mathbf{Y} - \mathbf{D}_R \mathbf{S} \mathbf{D}_T^H - \mathbf{C}\|_F^2 \end{aligned} \quad (6)$$

where $\mathbf{Z}_1, \mathbf{Z}_2 \in \mathbf{C}^{N_R \times N_T}$ are the Lagrange multipliers. ρ denotes ADMM's stepsize. According to standard ADMM, at the l th algorithmic iteration with $l = 0, 1, 2, \dots$, we need to solve the following subproblems

$$\mathbf{H}^{(l+1)} = \underset{\mathbf{H}}{\text{argmin}} L_1(\mathbf{H}, \mathbf{Y}^{(l)}, \mathbf{S}^{(l)}, \mathbf{C}^{(l)}, \mathbf{Z}_1^{(l)}, \mathbf{Z}_2^{(l)}) \quad (7)$$

$$\mathbf{Y}^{(l+1)} = \underset{\mathbf{Y}}{\text{argmin}} L_1(\mathbf{H}^{(l+1)}, \mathbf{Y}, \mathbf{S}^{(l)}, \mathbf{C}^{(l)}, \mathbf{Z}_1^{(l)}, \mathbf{Z}_2^{(l)}) \quad (8)$$

$$\mathbf{S}^{(l+1)} = \underset{\mathbf{S}}{\text{argmin}} L_1(\mathbf{H}^{(l+1)}, \mathbf{Y}^{(l+1)}, \mathbf{S}, \mathbf{C}^{(l)}, \mathbf{Z}_1^{(l)}, \mathbf{Z}_2^{(l)}) \quad (9)$$

$$\mathbf{C}^{(l+1)} = \underset{\mathbf{C}}{\text{argmin}} L_1(\mathbf{H}^{(l+1)}, \mathbf{Y}^{(l+1)}, \mathbf{S}^{(l+1)}, \mathbf{C}, \mathbf{Z}_1^{(l)}, \mathbf{Z}_2^{(l)}) \quad (10)$$

$$\mathbf{Z}_1^{(l+1)} = \mathbf{Z}_1^{(l)} + \rho (\Omega^\circ \mathbf{Y}^{(l+1)} - \mathbf{H}_\Omega) \quad (11)$$

$$\mathbf{Z}_2^{(l+1)} = \mathbf{Z}_2^{(l)} + \rho (\mathbf{Y}^{(l+1)} - \mathbf{D}_R \mathbf{S}^{(l+1)} \mathbf{D}_T^H - \mathbf{C}^{(l+1)}) \quad (12)$$

for the initialization $l = 0$, $\mathbf{H}^{(0)} = \mathbf{Z}_1^{(0)} = \mathbf{Z}_2^{(0)} = 0$. Then we derive closed form solutions for the above subproblems. To solve (7), We reformulate L_1 to L_2 as follows

$$L_2(\mathbf{H}) \triangleq \tau_H \|\mathbf{H}\|_* + \frac{\rho}{2} \left\| \mathbf{H} - \left(\mathbf{Y}^{(l)} - \frac{1}{\rho} \mathbf{Z}_1^{(l)} \right) \right\|_F^2 \quad (13)$$

Given the Lagrangian in (13), the solution of $\mathbf{H}$ in (7) can be obtained from the randomized SVD(i.e., Alg.1).

$$\mathbf{H}^{(l+1)} = \mathbf{U} \text{diag}(\{\text{sign}(\xi_i) \max(\xi_i, 0)\}_{1 \leq i \leq r}) \mathbf{V}^H \quad (14)$$

where $\mathbf{U} \in \mathbf{C}^{N_R \times r}$ and $\mathbf{V} \in \mathbf{C}^{N_T \times r}$ are the left and right singular vector matrices of the matrix $\mathbf{Y}^{(l)} - \frac{1}{\rho} \mathbf{Z}_1^{(l)}$, respectively. $\xi_i \triangleq \sigma_i - \tau/\rho$ with σ_i s denotes its r singular values.

Similarly, to derive closed form solutions of $\mathbf{Y}$ in (8), we reformulate L_1 to L_3 of $\mathbf{Y}$ as follows

$$\begin{aligned} L_3(\mathbf{Y}) \triangleq & \|\Omega^\circ \mathbf{Y} - \mathbf{H}_\Omega\|_F^2 + \frac{\rho}{2} \left\| \frac{1}{\rho} \mathbf{Z}_1^{(l)} + \mathbf{H}^{(l+1)} - \mathbf{Y} \right\|_F^2 \\ & + \frac{\rho}{2} \left\| \frac{1}{\rho} \mathbf{Z}_2^{(l)} + \mathbf{C}^{(l)} - \mathbf{Y} + \mathbf{D}_R \mathbf{S}^{(l)} \mathbf{D}_T^H \right\|_F^2 \end{aligned} \quad (15)$$

(15) can be equivalently expressed based on the Krockecker vectorization and the Hadamard element-wise property as

$$\begin{aligned} L_3(\mathbf{y}) \triangleq & \frac{1}{2} \|\mathbf{A} \mathbf{y} - \mathbf{h}_\Omega\|_2^2 + \frac{\rho}{2} \left\| \frac{1}{\rho} \mathbf{z}_1^{(l)} + \mathbf{h}^{(l+1)} - \mathbf{y} \right\|_2^2 \\ & + \frac{\rho}{2} \left\| \frac{1}{\rho} \mathbf{z}_2^{(l)} + \mathbf{c}^{(l)} - \mathbf{y} + \mathbf{B} \mathbf{s}^{(l)} \right\|_2^2 \end{aligned} \quad (16)$$

where $\mathbf{B} \triangleq \mathbf{D}_T^* \otimes \mathbf{D}_R$. $\mathbf{A} \triangleq \sum_{i=1}^{N_R} \text{diag}([\Omega]_i)^T \otimes \mathbf{E}_{ii}$, $[\Omega]_i$ denotes Ω 's i th row. $\mathbf{E}_{ii}$ can be obtained from the $N_R \times N_R$ all-zero matrix with inserting a unity value at its (i, i) th position. Meanwhile $\mathbf{Y}^{(l+1)} = \text{unvec}(\mathbf{y}^{(l+1)})$ and $\mathbf{y}$ can be obtained from

$$\begin{aligned} \mathbf{y}^{(L+1)} = & (\mathbf{A}^H \mathbf{A} + 2\rho \mathbf{I})^{-1} \\ & \left(\mathbf{z}_1^{(l)} + \rho \mathbf{h}^{(l+1)} + \mathbf{A}^H \mathbf{h}_\Omega + \mathbf{z}_2^{(l)} + \rho \mathbf{c}^{(l)} + \rho \mathbf{B} \mathbf{s}^{(l)} \right) \end{aligned} \quad (17)$$

To find $\mathbf{S}$ in (9), we reformulate L_1 to L_4 as follows

$$L_4(\mathbf{S}) \triangleq \frac{\rho}{2} \left\| \mathbf{D}_R^H \left(\frac{1}{\rho} \mathbf{Z}_2^{(l)} - \mathbf{C}^{(l)} + \mathbf{Y}^{(l+1)} \right) \mathbf{D}_T + \mathbf{S} \right\|_F^2 + \tau_S \|\mathbf{S}\|_1 \quad (18)$$

By performing vectorization and introducing $\tau'_s \triangleq \tau_S/\rho$, $\mathbf{s}$ can be obtained from

$$\mathbf{s}^{(l+1)} = \text{sign}(\text{Re}(\mathbf{v}^{(l+1)})) \circ \max(|\text{Re}(\mathbf{v}^{(l+1)})| - \tau'_s, 0) + \text{sign}(\text{Im}(\mathbf{v}^{(l+1)})) \circ \max(|\text{Im}(\mathbf{v}^{(l+1)})| - \tau'_s, 0) \quad (19)$$

where $\mathbf{v}^{(l+1)} \triangleq \text{vec}(\mathbf{V}^{(l+1)})$ with

$$\mathbf{V}^{(l+1)} \triangleq \mathbf{D}_R^H \left(\frac{1}{\rho} \mathbf{Z}_2^{(l)} - \mathbf{C}^{(l)} + \mathbf{Y}^{(l+1)} \right) \mathbf{D}_T \quad (20)$$

The solution of $\mathbf{C}$ in (10) is

$$L_5(\mathbf{C}) \triangleq \frac{\rho}{2} \left\| \frac{1}{\rho} \mathbf{Z}_2^{(L)} + \mathbf{Y}^{(l+1)} - \mathbf{D}_R \mathbf{S}^{(l+1)} \mathbf{D}_T^H - \mathbf{C} \right\|_F^2 + \frac{1}{2} \|\mathbf{C}\|_F^2 \quad (21)$$

which is strictly convex with respect to $\mathbf{C}$. So we get the closed solution set of $\mathbf{C}$ by deriving it and making it zero.

$$\mathbf{C}^{(l+1)} = \frac{\rho}{\rho + 1} \left(\mathbf{Y}^{(l+1)} - \mathbf{D}_R \mathbf{S}^{(l+1)} \mathbf{D}_T^H - \frac{1}{\rho} \mathbf{Z}_2^{(l)} \right) \quad (22)$$

Alg.3 summarizes the ADMM steps which constitute the proposed mmWave massive MIMO channel estimation algorithm. After we set the number of algorithmic iterations $\mathbf{I}_{\max}$, we can get the estimated MIMO channel matrix $\mathbf{H}^{(\mathbf{I}_{\max})}$.

Algorithm 3 Proposed MIMO Channel Estimation Algorithm

Input: $\mathbf{H}_\Omega, \Omega, \mathbf{D}_R, \mathbf{D}_T, \rho, \tau_H, \tau_S$ and $\mathbf{I}_{\max}$

Output: $\mathbf{H}^{(\mathbf{I}_{\max})}$

- 1: Initialization: $\mathbf{H}^{(0)} = \mathbf{S}^{(0)} = \mathbf{C}^{(0)} = \mathbf{Z}_2^{(0)} = \mathbf{Z}_1^{(0)} = \mathbf{0}$
 - 2: **for** $l = 0, 1, \dots, \mathbf{I}_{\max} - 1$ **do**
 - 3: Update $\mathbf{H}^{(l+1)}$ using (14) with Alg.1.
 - 4: Update $\mathbf{Y}^{(l+1)} = \text{unvec}(\mathbf{y}^{(l+1)})$ using (17).
 - 5: Update $\mathbf{S}^{(l+1)} = \text{unvec}(\mathbf{s}^{(l+1)})$ using (19).
 - 6: Update $\mathbf{C}^{(L+1)}$ using (22).
 - 7: Update $\mathbf{Z}_1^{(l+1)}$ and $\mathbf{Z}_2^{(l+1)}$ using (11) and (12) respectively.
 - 8: **end for**
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C. Computational Complexity

The computational complexity of Alg.3 mainly depends on the number of antennas N_T and N_R , as well as the number of iterations $\mathbf{I}_{\max}$. In the third step of Alg.3, using SVD to calculate σ_i s to update $\mathbf{H}^{(l+1)}$ is the main source of complexity. Specifically, In the first step of Alg.1, the complexity of generating random matrix Ω is $\mathcal{O}(N_T(k+s))$. The second step of matrix multiplication $\mathbf{A}\Omega$ will produce a complexity of $\mathcal{O}(N_R N_T(k+s))$. Then the LU factorization of the matrix $\mathbf{A}\Omega$ will produce a complexity of $\mathcal{O}(N_R(k+s)^2)$ [17]. The power iteration from step 3 to step 5 is the main source of complexity, it will produce a complexity of $\mathcal{O}(1/3 N_R p^3(k+s)^2)$ with step 2. In the step 7, the QR decomposition will produce a complexity of $\mathcal{O}(N_R p^2(k+s)^2)$. The complexity of step 8

and step 9 are mainly as follows $\mathcal{O}(N_T^2(k+s))$. Through the above complexity analysis, since p and s are usually integers less than 5, the value of k is related to the number of antennas, but $k \ll \text{rank}(\mathbf{A})$. Due to the number of base station antennas is typically increased to several times the number of users in massive MIMO systems [18], the complexity of step 3 in Alg.3 is $\mathcal{O}(1/3 N_R p^3(k+s)^2)$, which is more effective than complete singular value decomposition methods(e.g., ADMM-based algorithm [8], which is $\mathcal{O}(N_R^2 N_T)$).

IV. SIMULATION RESULTS

We consider the example of massive MIMO systems in the form of ULAs with a 90 GHz mmWave channel. The azimuthal AOA and deviation are generated according to the Laplacian distribution with the standard deviation of 50° . As for benchmark CSI estimation techniques, the following algorithms are chosen as the simulation comparison.

- OMP [4]
- TSSR [5]
- ADMM [8]
- SVT [9]
- VAMP [10]

Note that OMP and VAMP exploit only the sparsity of the channel matrix, while SVT capitalizes only on its low-rank property. TSSR, ADMM and the proposed algorithm exploit both properties. For OMP and VAMP, the parameter representing the channel sparsity was set to the number of paths(N_p). For SVT, the threshold $\tau = \rho \|\mathbf{H}_\Omega\|$ with $\rho = \frac{3M}{N_R N_T}$. For ADMM and the proposed Algorithm 4, we used $\tau_H = \rho \|\mathbf{H}_\Omega\|$ with $\rho = 0.005$ and $\tau_S = \frac{0.1}{1-10 \log(\sigma_n^2)}$.

We compared the performance of channel estimation algorithms both in terms of the normalized mean square error(NMSE) and achievable spectral efficiency(ASE) in bits/second/hertz. For the algorithm ADMM, SVT and the proposed that based on matrix completion, the convergence of the algorithms was investigated under different training sequence lengths as well. Note that $\hat{\mathbf{H}}$ represents the estimation for the real channel $\mathbf{H}$ and it's given by $\mathbf{H}^{(\mathbf{I}_{\max})}$. NMSE can be calculated by the following formula

$$\text{NMSE} \triangleq \varepsilon \left\{ \|\hat{\mathbf{H}} - \mathbf{H}\|_F^2 / \|\mathbf{H}\|_F^2 \right\} \quad (23)$$

ASE can be calculated by

$$\text{ASE} \triangleq \varepsilon \left\{ \log_2 \det \left(\mathbf{I}_{N_R} + (N_T N_R (\sigma_n^2 + \text{NMSE}))^{-1} \mathbf{H} \mathbf{H}^H \right) \right\} \quad (24)$$

For these two expressions, we obtained the expectations from average of 50 independent Monte Carlo realizations. The simulation results demonstrate that the proposed algorithm outperforms the comparison algorithms OMP, TSSR, VAMP and ADMM in terms of the NMSE performance and ASE for short training sequences T under a low SNR environment.

Fig.1 shows that after about 40 iterations, all three algorithms converge, which provides a reference for us to choose the value of $\mathbf{I}_{\max}$. When the length of the training sequences is increased to 1200, there is little difference between the performance of the proposed algorithm and ADMM, because

the length of the training sequences can improve the accuracy of channel estimation.

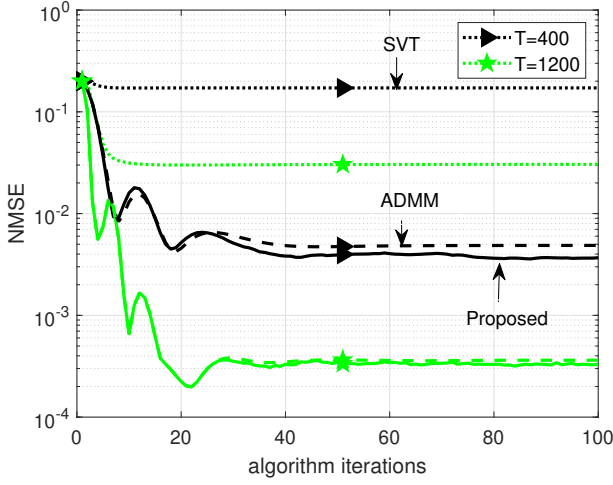


Fig. 1. NMSE w.r.t. algorithmic iterations and different T values for a 64×64 channel with 30dB transmit SNR.

In Fig.2, due to the discretization error of the AoA, the performance of OMP can not be improved over the training sequences T or SNR ratio. VAMP can not accurately recover the CSI with short training sequences as well because it's based on the calculation of the statistical information of the sparse signal, which is difficult to acquire with small T . For TSSR, it first uses SVT to recover the channel matrix and then uses it as input to VAMP. Thus it can't recover the CSI either. The proposed algorithm and ADMM can achieve good convergence effect because they both use the computational framework of ADMM. However, the proposed algorithm can get more accurate estimation result with the use of randomized SVD.

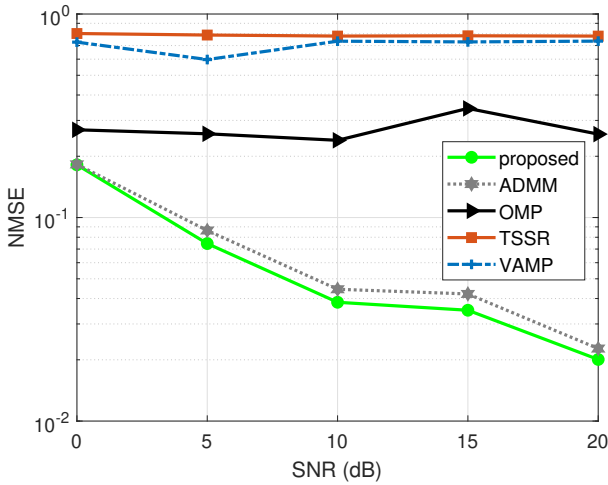


Fig. 2. NMSE w.r.t. transmit SNR for a 128×32 channel with $N_p = 2$ and $T = 400$.

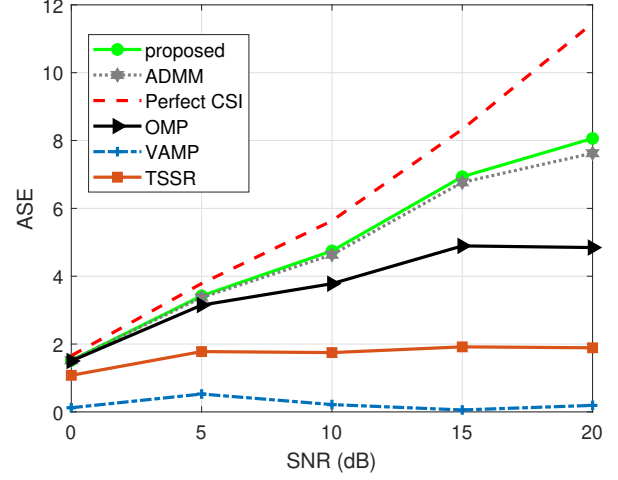


Fig. 3. ASE w.r.t. transmit SNR for a 128×64 channel with $N_p = 2$ and $T = 400$.

Fig.3 presents the ASE performance of the above all algorithms with $N_p = 2$ and $T = 400$ in a 64×64 channel. We also depict the result of perfect CSI as a comparison. Clearly, the proposed algorithm can be closer to the ASE of perfect CSI. The algorithm VAMP and TSSR can't achieve satisfactory spectral efficiency with the short training sequences.

V. CONCLUSION

In this work, a novel channel algorithm based on matrix completion and randomized SVD is proposed by utilizing the sparse and low-rank features of the mmWave channel. We first establish a low-rank matrix recovery problem according to the matrix completion theory. Then the beamspace decomposition model of the channel matrix is introduced to establish a dual objective convex optimization problem. Finally, in the process of solving, we mainly use the ADMM method and introduce randomized SVD to accelerate the algorithm. The simulation results show that the proposed algorithm can show more accurate estimation performance with shorter training sequences length, and the spectral efficiency is closer to that of perfect CSI. As future work, it's significant to study the channel estimation algorithm for wideband mmWave transmission and the algorithm with strong robustness under different antenna arrays.

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